Project Two Design Defense

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CS-370: Current and Emerging Trends in Computer Science

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Human Learning vs. Machine Learning

Humans use multiple skills to solve problems, including abstract reasoning, pattern recognition, and heuristics, a technique that uses trial and error and past experiences to make quick judgements. A human player in a treasure maze would use these skills to determine the best strategy to achieve the objective of getting through the maze and finding the treasure. The program's AI agent, on the other hand, learned how to solve the problem of finding the treasure by a process called value-based learning. This form of machine learning involves estimating the value of every possible action the agent can take to choose a valid move that will earn the most points until it solves the maze. This process repeats itself until the agent produces the best gameplay strategy.

Using Algorithms to Solve Complex Problems

The implementation of deep Q-learning for this project's treasure maze game was straight-forward. First, certain libraries such as TensorFlow needed to be imported in order for the program to function properly. After that, the training environment needed to be created. This involved creating a set of rules for the agent to follow (i.e. what is considered a valid or invalid move), determining the rewards and penalties for a given move, defining the winning and losing conditions of the game, and printing information about how the agent is progressing through the training process as the output. Finally, attempts were made to test the agent within the

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environment. Unfortunately, when I ran the program, it gave me a runtime error. I tried to fix the issue multiple times but to no avail.

References

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